

Applied Diagnostic: Digikala Sales Drop

“Digikala sales dropped 20%. What metrics would you investigate?”

A worked diagnostic that applies the North Star metric tree from the companion case study to a hypothetical sales-drop scenario.

0. Validate the data

Before anything else, confirm the drop is real and not a tracking or pipeline issue causing under-reporting. If the reporting itself is broken, every downstream conclusion is worthless.

1. Clarify the definition

What does “sales” mean here — GMV or revenue? The two point to different root causes (order volume vs. take rate / cost structure), so this has to be settled before anything else.

2. Segment the drop

Before forming hypotheses, narrow down where the drop is concentrated:

- Digikala's own sales or third-party sellers?
- A specific product category?
- A specific city or region?
- A specific user cohort?
- Web or app?
- Sudden or gradual? Over what time window?

3. Categorize likely causes (qualitative)

External

- A competitor launching a feature that pulled users away.
- Unusual market or pricing volatility.
- A competitor's sale or discount campaign.
- A new competitor entering the market.
- If the window is long: shifting user behavior or market trends.

Internal — company-level

- Did a discount or campaign end during this window?
- Were data pipelines and infrastructure functioning correctly?
- Was there an operational or fulfillment issue on Digikala's or a third-party seller's side?

Internal — team-level

- Any recent UI/visual changes, new feature launches, or funnel-flow changes that line up with the timing of the drop.

4. Decompose the drop through the metric tree

Rather than leaving the causes above as qualitative guesses, I quantify the drop using the North Star metric tree (see the companion Digikala metric-framework case study) to see exactly where the 20% is coming from before investigating why.

GMV vs. completed orders

Did GMV fall because AOV dropped, or because the number of completed orders dropped?

Flat AOV + fewer orders → demand/funnel issue. Falling AOV + flat orders → pricing or product-mix issue.

Completed orders

Total orders $\times$ (1 – cancellation rate) — did cancellation rate rise, or did total orders fall?

A rising cancellation rate points to an operational or seller-quality issue (company-level cause).

Total orders

Active users $\times$ repeat purchase rate — which one moved?

A falling repeat purchase rate means existing users are buying less often — often a satisfaction or competitor-driven signal.

Active users

New users vs. returning users — which one dropped?

Falling new users → acquisition issue (budget, competitor campaign). Falling returning users → retention issue.

New users

Visitors $\times$ conversion rate — did traffic drop, or did conversion drop?

This connects directly to the funnel analysis in step 5.

This step turns a vague hypothesis like “maybe it's the competitor, maybe it's operational” into something specific and testable — for example: “70% of the drop traces to a falling repeat purchase rate among existing users, not new traffic.” That narrows step 3's hypotheses to the ones actually worth investigating.

5. Funnel analysis

Starting from the landing page through order delivery, I trace where drop-off is concentrated — focused specifically on the stage the metric tree pointed to in step 4, rather than re-auditing the entire funnel from scratch.

6. Validate the hypothesis

Once the data points to a leading hypothesis, I validate it before acting on it, choosing the method based on available time and the cost of being wrong:

- Compare against historical data, especially year-over-year, to rule out seasonality.
- Run an A/B test.
- Interview a sample of affected users.
- Build a lightweight test (e.g. a fake-door button) to isolate the signal quickly.

Note: This is a structured reasoning exercise, not an analysis of real Digikala data. It demonstrates how the metric tree from the companion case study can be applied to diagnose a live business problem.